

# Supplementary Information for

## Quantization fragility under image corruption is recipe-dependent: paired evidence from INT8 and FP8 object detectors

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### S1 Purpose and scope

This supplement documents the frozen protocol details, estimands, corruption parameters, calibration schedules, and artifact contract supporting the main article. All reported values are generated deterministically from retained ledgers. Section S8 additionally reports per-family  $\Delta E$  diagnostics that support—but do not extend—the main-text claims.

### S2 Estimands

For dataset  $d$ , model  $m$ , corruption condition  $c$ , and a pair of treatments (A, B) sharing the same evaluator:

$$G_0 = \text{AP}_A(\text{J95}) - \text{AP}_B(\text{J95}), \quad (\text{S1})$$

$$G_{\text{corr}} = \overline{\text{AP}_A(c) - \text{AP}_B(c)} \quad (\text{equal weight over the 12 cells}), \quad (\text{S2})$$

$$\Delta E = G_{\text{corr}} - G_0. \quad (\text{S3})$$

J95 denotes the codec-matched clean control: the J95 clean control and every corrupted input are terminally encoded as JPEG quality 95 with chroma subsampling disabled, so the clean control and the corrupted inputs differ only in the named corruption transform, not in container encoding; the separate original-clean cell keeps the dataset-native bytes; it checks control fidelity and anchors the clean-accuracy comparisons behind the RetinaNet collapse and recovery claims, while the paired interaction uses the J95 basis.  $\Delta E > 0$  means the A-minus-B gap widens under corruption;  $\Delta E < 0$  means it narrows. A narrowing gap near a shared accuracy floor is not read as robustness: absolute corrupted AP is reported alongside every interaction (main text, §4.1–§4.3).

### S3 Frozen condition schedule

Each dataset×model block evaluates 14 conditions on identical image ordering for quantized arms (13 for the RetinaNet FP32 and INT8-legacy arms that predate codec control):

- clean (original bytes) and J95 (JPEG-95 clean control);
- Gaussian noise:  $\sigma \in \{8, 20, 40\}$ ;
- motion blur: kernel lengths  $\{5, 13, 23\}$ ;
- synthetic fog: (strength, radius)  $\in \{(0.18, 6), (0.38, 12), (0.62, 20)\}$ ;
- JPEG compression: qualities  $\{70, 40, 15\}$ .

Corruption parameters and the JPEG-95 terminal encoding are frozen in versioned configuration `corruptions.json`, whose SHA-256 is recorded as `generator_config.sha256` in the corruption-aware and codec-control calibration manifests; the frozen protocol file `nn_paired_protocol.v1_20260924` fixes the condition list and evaluation contract on top of that generator. The same transform code produces both the evaluation caches and the corruption-aware calibration images.

## S4 Treatments and arm registry

Table S1: RetinaNet arm registry. “Calibration” lists the input distribution seen by entropy calibration; “sites” lists the quantized region.

| Arm                   | Precision | Quantized sites                                | Calibration input                                 |
|-----------------------|-----------|------------------------------------------------|---------------------------------------------------|
| FP32                  | FP32      | none                                           | —                                                 |
| FP8-matched           | FP8       | full graph                                     | clean, matched preprocessing                      |
| INT8-legacy           | INT8      | full graph                                     | mismatched preprocessing (diagnostic)             |
| INT8-matched          | INT8      | full graph                                     | clean, 512 images                                 |
| INT8-q95calib         | INT8      | full graph                                     | J95 re-encodes, 512 images                        |
| INT8-selective        | INT8      | all except <code>/head/regression_head/</code> | clean, 512 images                                 |
| INT8-sel-q95calib     | INT8      | selective                                      | J95 re-encodes, 512 images                        |
| INT8-corruptcalib     | INT8      | full graph                                     | fold-1 corruptions (noise, JPEG), 512 images      |
| INT8-sel-corruptcalib | INT8      | selective                                      | in-family corruptions, 512 images                 |
| INT8-cc2calib         | INT8      | full graph                                     | fold-2 corruptions (fog, motion blur), 512 images |
| INT8-sel-cc2calib     | INT8      | selective                                      | fold-2 corruptions, 512 images                    |
| INT8-wonly            | INT8      | full graph; head <i>acts</i> . fp16            | clean, 512 images                                 |
| INT8-aonly            | INT8      | full graph; head <i>weights</i> fp16           | clean, 512 images                                 |
| FP8-legacy            | FP8       | full graph                                     | pre-study build (context only)                    |

The W-only and A-only arms are post-hoc QDQ-graph rewires of the frozen INT8-matched graph (Section 3.7 of the main text): under the TensorRT `int8-entropy` build policy every non-quantized operation executes in fp16, so the restored operand computes in fp16 rather than literal fp32, and the bypassed head weight initializers are themselves fp16 tensors.

Table S2: FCOS-R50-FPN replication arm registry. The selective exclusion regex `~/head/regression_head/` covers the regression tower and the `bbox_ctrness` output, so the selective arm also dequantizes centerness—because the score is  $\sqrt{\sigma(\text{cls}) \cdot \sigma(\text{ctr})}$ , the intervention is not treatment-identical to the RetinaNet selective arm, which touches regression outputs only. Calibration identical to the RetinaNet matched recipe.

| Arm            | Precision | Quantized sites                          | Calibration input            |
|----------------|-----------|------------------------------------------|------------------------------|
| FP32           | FP32      | none                                     | —                            |
| FP8-matched    | FP8       | full graph                               | clean, matched preprocessing |
| INT8-matched   | INT8      | full graph                               | clean, 512 images            |
| INT8-selective | INT8      | all except regression tower + centerness | clean, 512 images            |

All INT8/FP8 engines are built by TensorRT 11.1.0.106 from ONNX graphs quantized by NVIDIA ModelOpt with entropy calibration (512-image budget; the study-built registries record version 0.45.0 and the pre-study YOLO11 ONNX registries predate that field; the diagnostic sweeps vary the estimator and include arms marked “128-img” in Table 4 of the main text, which use the earlier 128-image subset) with a 4,096 MiB builder workspace. Builder policy is recorded per engine and differs by program: the YOLO11 engines were built by `trtexec` before the frozen contract and carry TensorRT 11.1 defaults (TF32 enabled); the RetinaNet quantization arms built for this study use the TensorRT Python builder with TF32 disabled, while the RetinaNet FP32 reference and the FP8/INT8-legacy arms are pre-study `trtexec` builds with TF32 enabled (the  $\Delta E$  contrasts only join quantized arms built under the same policy); the four FCOS arms, FP32 included, are TensorRT Python builder builds with TF32 disabled, so the FCOS block is internally policy-matched but not builder-identical to the pre-study YOLO11 engines. Calibration and evaluation images are disjoint. Checkpoints are one per dataset×model cell, and their selection provenance is disclosed: the KITTI and VOC YOLO11n/m/x models were fine-tuned from Ultralytics pretrained weights under the pilot program (`*_train_v1` runs, dataset YAMLs: KITTI `val: images/val`, VOC `val: images/test2007`), with the best checkpoint taken on the same partition that serves as the evaluation universe; COCO YOLO11 uses the official pretrained checkpoint without a dataset fine-tune; the torchvision RetinaNet-R50-FPN-v2 model is fine-tuned per dataset for 100 epochs at 640-pixel input (learning rate  $10^{-4}$ , weight decay  $10^{-4}$ , deterministic seed 20260807), selecting the best checkpoint by minimum validation loss on the same evaluation partition. FCOS-R50-FPN is trained under the same profile and checkpoint rule. All four RetinaNet/FCOS selected checkpoints are early picks: run-log epochs 21 (KITTI) and 5 (VOC) for RetinaNet, 14 (KITTI) and 10 (VOC) for FCOS (epoch counts here are the run log’s 1-indexed values; the checkpoints store the same point 0-indexed). The KITTI/VOC YOLO11 models instead select late best-fitness checkpoints (run epochs 65–100; recorded in `phase_a_arms/training/yolo11.best_fitness_epochs.json`) under Ultralytics’ own rule, and the COCO YOLO11 models are the official pretrained weights with no study-side selection. Both FCOS runs completed all 100 scheduled epochs. On VOC the FCOS training loss diverged to NaN at epoch 70 (validation

loss stayed finite through epoch 92, then also diverged); the epoch-10 selection point predates the divergence by 60 epochs, and no intervening epoch beat its validation minimum, so the divergence did not affect which checkpoint was exported. The training manifest records the schedule and selection rule but not the loss history; the per-epoch training/validation loss curves are shipped in the support package (`phase_b_results/training/` for RetinaNet, `phase_h_fcos_results/training/` for FCOS), and the selected epochs match the checkpoints’ stored epoch fields (raw run logs are not redistributed; each curve record carries its source log’s SHA-256). Per-run training manifests are hash-bound; the evaluation-partition overlap is a disclosed selection caveat (see the main text’s limitations). The frozen condition schedule is 14 conditions per arm; the RetinaNet FP32 and legacy arms predate codec control and carry 13.

## S5 Inference and evaluation contract

YOLO11 blocks: 640-pixel letterbox input, confidence threshold 0.001, class-wise NMS at IoU 0.7, at most 300 detections. RetinaNet blocks: 640-pixel letterbox input, confidence threshold 0.05, batched NMS at IoU 0.5, at most 300 detections. FCOS blocks: 640-pixel letterbox input, confidence threshold 0.05 on the centerness-weighted score, at most 1,000 pre-NMS candidates per feature level, batched NMS at IoU 0.5, at most 300 detections—the RetinaNet constants reused post hoc and frozen for FCOS at the RetinaNet values, and not torchvision’s FCOS defaults (score threshold 0.2, NMS IoU 0.6, 100 detections per image); the decoder is a reimplement of torchvision’s post-processing, validated detection-for-detection against torchvision’s own pipeline on 32 seeded synthetic head-output draws (`phase_h_fcos_results/decoder_validation.json`: 9,600 detections, maximum box and score deviation exactly zero). The check saturates the 300-detection cap on every draw, uses a square 640-pixel input at scale 1.0, and covers the 8-class KITTI head only—it validates the post-processing arithmetic, not letterbox un-mapping, sparse-output behavior, or end-to-end engine parity. All evaluations use the COCO protocol via `pycocotools` with `maxDets=100`; VOC and KITTI annotations were converted once to COCO format and frozen. Decoder constants are listed per family in the paragraph above—identical for RetinaNet and FCOS—and held fixed within each; cross-family comparisons are comparisons of recipe bundles, never of formats alone.

## S6 Corruption-aware calibration schedule

The intervention calibration sets reuse the same 512 images as the matched clean-calibration manifests. Each image is deterministically assigned—via a SHA-256-keyed hash of its identifier—to exactly one of six in-family cells, Gaussian noise or JPEG at severities 1, 3, or 5 (approximately 77–102 images per cell). Assigned corruptions use the frozen transform parameters above, and the outputs are terminally encoded at JPEG quality 95, matching the evaluation input contract. In this first fold fog and motion blur never appear in calibration and form the held-out split. A complementary second fold (schedule `nn-corruptcalib-fold2-v1`, frozen before any of its runs) assigns the same images to fog or motion blur at severities 1, 3, or 5 (75–95 images per cell) and holds out Gaussian noise and JPEG. Its pre-specified rules are (i) a held-out change from INT8-matched whose interval excludes zero (either direction counts; the main text reads it through the directional fold-1 criterion) and (ii) a positive held-out  $\Delta E$  with an interval excluding zero, against both INT8-matched and INT8-q95calib.

The codec-matched control manifests encode the same 512 images at JPEG quality 95 with no named corruption, isolating codec adaptation from corruption coverage.

## S7 Per-arm intervention contrasts

Table S3 gives the full per-arm contrast set underlying Section 4.4 of the main text, and Fig. S1 groups the same contrasts by calibration arm and evaluation partition.

The additive  $\Delta E$  decomposition is itself scale-sensitive (a uniform level shift on a degraded arm appears as a corruption-specific gain), so we run a retention-scaled check: each fold’s clean-control change is scaled by the matched arm’s AP retention on that fold’s held-out families (0.63 for fold 1 and 0.82 for fold 2 on KITTI; 0.67 and 0.78 on VOC), predicting held-out changes of +0.98 and -2.22 on KITTI and +0.49 and +0.82 on VOC against the observed +1.07, -2.02, +0.02, and +1.66. Under this reading both KITTI cells are largely scale effects, VOC fold 1 still shows no corruption-specific benefit, and only VOC fold 2 exceeds its prediction materially (+1.66 vs. +0.82); anchoring on the codec arm gives the same qualitative answer (KITTI fold-1 residual -0.23 AP).

The fold-1 primary estimand also covers the selective-tier held-out gain over its clean-calibrated counterpart (sel-corruptcalib minus sel) is -0.58 [-0.80, -0.36] on KITTI—significantly negative, under the development-contamination caveat of the main text—and -0.01 [-0.17, +0.13] on VOC. On KITTI fold 2 the selective arm is significantly lifted where the full-INT8 arm is harmed (J95 +1.01 [+0.37, +1.72]; held-out +0.66 [+0.33, +0.92]), but neither lift survives the codec anchor (sel-cc2calib minus sel-q95calib: J95 +0.75 [-0.04, +1.56]; held-out +0.25 [-0.09, +0.52]). Against FP8-matched, the VOC fold-2 selective arm’s held-out residual (+0.08 [-0.06, +0.23])

Table S3: RetinaNet intervention and codec-control contrasts (A minus B, AP points; plug-in point with 95% paired-bootstrap interval, 2,000 common-image resamples). J95 clean: codec-control clean cell; corrupt mean: 12 cells; in-family: Gaussian noise and JPEG (the fold-1 calibrated families); held-out: fog and motion blur (never calibrated in fold 1); for the q95-calibration rows, which see clean images only, these labels mark the evaluation partition rather than exposure;  $\Delta E_{\text{held-out}}$  = held-out difference minus J95 clean difference, and  $\Delta E_{\text{in-family}}$  is defined analogously; because every row here is an intervention-minus-baseline contrast (the  $\Delta E$  definition of the main text), positive  $\Delta E$  shifts the difference toward the first-named arm. For rows anchored to a clean-calibrated INT8 arm, a positive  $\Delta E$  is a corruption-specific intervention benefit; for rows anchored to FP8-matched—which remain deep level deficits—positive values describe gap narrowing under corruption, not a benefit. Fold-2 rows swap the family roles: their in-family is fog and motion blur and their held-out is Gaussian noise and JPEG. \* interval excludes zero.

| Contrast (A – B)                               | Dataset | J95 clean                | Corrupt mean             | In-family                | Held-out                 | $\Delta E_{\text{in-family}}$ | $\Delta E_{\text{held-out}}$ |
|------------------------------------------------|---------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------------|------------------------------|
| INT8-corruptcalib – INT8-matched               | KITTI   | +1.56* [+1.02, +2.10]    | +1.34* [+1.17, +1.48]    | +1.61* [+1.37, +1.79]    | +1.07* [+0.87, +1.26]    | +0.05 [–0.54, +0.59]          | –0.50 [–1.01, +0.03]         |
| INT8-corruptcalib – INT8-matched               | VOC     | +0.73* [+0.40, +1.01]    | +0.64* [+0.52, +0.76]    | +1.27* [+1.12, +1.42]    | +0.02 [–0.13, +0.16]     | +0.54* [+0.24, +0.88]         | –0.71* [–1.02, –0.39]        |
| INT8-q95calib – INT8-matched                   | KITTI   | –0.34 [–0.93, +0.35]     | +0.04 [–0.10, +0.22]     | –0.02 [–0.23, +0.21]     | +0.11 [–0.05, +0.33]     | +0.31 [–0.35, +0.90]          | +0.45 [–0.24, +1.04]         |
| INT8-q95calib – INT8-matched                   | VOC     | +0.20 [–0.07, +0.48]     | –0.17* [–0.26, –0.09]    | –0.23* [–0.37, –0.11]    | –0.11* [–0.228, –0.002]  | –0.44* [–0.73, –0.14]         | –0.31* [–0.62, –0.02]        |
| INT8-corruptcalib – INT8-q95calib              | KITTI   | +1.90* [+1.28, +2.54]    | +1.29* [+1.08, +1.45]    | +1.64* [+1.33, +1.86]    | +0.95* [+0.73, +1.15]    | –0.26 [–0.89, +0.30]          | –0.94* [–1.60, –0.33]        |
| INT8-corruptcalib – INT8-q95calib              | VOC     | +0.52* [+0.22, +0.80]    | +0.81* [+0.70, +0.93]    | +1.50* [+1.35, +1.66]    | +0.13 [–0.02, +0.28]     | +0.98* [+0.70, +1.31]         | –0.39* [–0.69, –0.07]        |
| INT8-sel-corruptcalib – INT8-selective         | KITTI   | –0.50 [–0.96, +0.03]     | –0.50* [–0.65, –0.36]    | –0.41* [–0.63, –0.24]    | –0.58* [–0.80, –0.36]    | +0.09 [–0.43, +0.52]          | –0.08 [–0.68, +0.41]         |
| INT8-sel-corruptcalib – INT8-selective         | VOC     | +0.63* [+0.33, +0.94]    | +0.61* [+0.49, +0.72]    | +1.23* [+1.08, +1.39]    | –0.01 [–0.17, +0.13]     | +0.60* [+0.29, +0.92]         | –0.65* [–0.97, –0.35]        |
| INT8-sel-q95calib – INT8-selective             | KITTI   | +0.26 [–0.15, +0.74]     | +0.36* [+0.23, +0.49]    | +0.41* [+0.21, +0.61]    | +0.31* [+0.13, +0.54]    | +0.15 [–0.39, +0.58]          | +0.05 [–0.48, +0.57]         |
| INT8-sel-q95calib – INT8-selective             | VOC     | +0.16 [–0.10, +0.43]     | –0.12* [–0.21, –0.04]    | –0.14* [–0.27, –0.01]    | –0.11 [–0.22, +0.01]     | –0.30* [–0.60, –0.02]         | –0.27 [–0.56, +0.02]         |
| INT8-sel-corruptcalib – INT8-sel-q95calib      | KITTI   | –0.76* [–1.35, –0.17]    | –0.85* [–1.05, –0.69]    | –0.82* [–1.08, –0.59]    | –0.89* [–1.16, –0.65]    | –0.06 [–0.65, +0.54]          | –0.13 [–0.83, +0.51]         |
| INT8-sel-corruptcalib – INT8-sel-q95calib      | VOC     | +0.47* [+0.16, +0.76]    | +0.73* [+0.61, +0.85]    | +1.37* [+1.22, +1.54]    | +0.09 [–0.07, +0.25]     | +0.90* [+0.61, +1.22]         | –0.38* [–0.68, –0.08]        |
| INT8-sel-corruptcalib – FP8-matched            | KITTI   | –0.23 [–1.12, +0.86]     | –0.67* [–0.91, –0.44]    | –0.03 [–0.37, +0.31]     | –1.30* [–1.61, –1.00]    | +0.20 [–0.74, +0.92]          | –1.08* [–2.18, –0.16]        |
| INT8-sel-corruptcalib – FP8-matched            | VOC     | +0.21 [–0.06, +0.49]     | –0.35* [–0.46, –0.24]    | –0.19* [–0.32, –0.05]    | –0.51* [–0.66, –0.38]    | –0.40* [–0.67, –0.12]         | –0.72* [–1.00, –0.44]        |
| INT8-corruptcalib – FP8-matched                | KITTI   | –19.51* [–20.90, –18.04] | –15.04* [–15.96, –14.17] | –16.74* [–17.85, –15.66] | –13.34* [–14.13, –12.57] | +2.77* [+1.98, +3.48]         | +6.17* [+5.12, +7.10]        |
| INT8-corruptcalib – FP8-matched                | VOC     | –2.31* [–2.68, –1.98]    | –2.13* [–2.30, –1.97]    | –2.13* [–2.33, –1.93]    | –2.12* [–2.32, –1.96]    | +1.77 [–1.10, +0.49]          | +0.18 [–0.12, +0.50]         |
| INT8-cc2calib – INT8-matched (fold 2)          | KITTI   | –2.71* [–3.38, –1.93]    | –1.69* [–1.88, –1.44]    | –1.35* [–1.56, –1.04]    | –2.02* [–2.31, –1.73]    | +1.36* [+0.61, +2.16]         | +0.69 [–0.10, +1.37]         |
| INT8-cc2calib – INT8-matched (fold 2)          | VOC     | +1.04* [+0.72, +1.34]    | +1.26* [+1.15, +1.37]    | +0.86* [+0.72, +0.99]    | +1.66* [+1.52, +1.83]    | –0.19 [–0.49, +0.13]          | +0.62* [+0.30, +0.96]        |
| INT8-cc2calib – INT8-q95calib (fold 2)         | KITTI   | –2.37* [–2.90, –1.74]    | –1.73* [–1.96, –1.47]    | –1.47* [–1.68, –1.18]    | –2.00* [–2.32, –1.69]    | +0.91* [+0.28, +1.47]         | +0.37 [–0.28, +0.90]         |
| INT8-cc2calib – INT8-q95calib (fold 2)         | VOC     | +0.84* [+0.54, +1.12]    | +1.43* [+1.32, +1.55]    | +0.97* [+0.82, +1.12]    | +1.89* [+1.75, +2.06]    | +0.13 [–0.15, +0.43]          | +1.06* [+0.76, +1.38]        |
| INT8-sel-cc2calib – INT8-selective (fold 2)    | KITTI   | +1.01* [+0.37, +1.72]    | +0.62* [+0.42, +0.83]    | +0.59* [+0.39, +0.86]    | +0.66* [+0.33, +0.92]    | –0.42 [–1.08, +0.24]          | –0.35 [–1.09, +0.28]         |
| INT8-sel-cc2calib – INT8-selective (fold 2)    | VOC     | +0.63* [+0.34, +0.94]    | +1.06* [+0.96, +1.18]    | +0.63* [+0.48, +0.76]    | +1.50* [+1.35, +1.66]    | –0.01 [–0.33, +0.29]          | +0.86* [+0.55, +1.18]        |
| INT8-sel-cc2calib – INT8-sel-q95calib (fold 2) | KITTI   | +0.75 [–0.04, +1.56]     | +0.27* [+0.08, +0.43]    | +0.28* [+0.03, +0.52]    | +0.25 [–0.09, +0.52]     | –0.47 [–1.25, +0.33]          | –0.50 [–1.27, +0.33]         |
| INT8-sel-cc2calib – INT8-sel-q95calib (fold 2) | VOC     | +0.48* [+0.18, +0.75]    | +1.19* [+1.07, +1.30]    | +0.74* [+0.59, +0.87]    | +1.64* [+1.49, +1.80]    | +0.26 [–0.03, +0.56]          | +1.16* [+0.87, +1.49]        |
| INT8-sel-cc2calib – FP8-matched (fold 2)       | KITTI   | +1.29* [+0.64, +2.03]    | +0.45* [+0.20, +0.71]    | –0.13 [–0.44, +0.20]     | +1.04* [+0.64, +1.40]    | –1.42* [–2.14, –0.76]         | –0.25 [–0.93, +0.36]         |
| INT8-sel-cc2calib – FP8-matched (fold 2)       | VOC     | +0.22 [–0.06, +0.48]     | +0.11 [–0.01, +0.22]     | +0.13 [–0.02, +0.26]     | +0.08 [–0.06, +0.23]     | –0.08 [–0.36, +0.20]          | –0.14 [–0.40, +0.15]         |
| INT8-cc2calib – FP8-matched (fold 2)           | KITTI   | –23.78* [–25.30, –22.13] | –18.07* [–19.07, –17.00] | –15.76* [–16.63, –14.86] | –20.37* [–21.58, –19.14] | +8.02* [+7.05, +8.94]         | +3.41* [+2.51, +4.23]        |
| INT8-cc2calib – FP8-matched (fold 2)           | VOC     | –1.99* [–2.34, –1.66]    | –1.51* [–1.68, –1.35]    | –1.28* [–1.47, –1.12]    | –1.74* [–1.94, –1.53]    | +0.70* [+0.41, +1.01]         | +0.25 [–0.04, +0.57]         |

is the only cell whose fold-1-style null-seeking interval includes zero; on KITTI the same contrast excludes zero favorably (+1.04 [+0.64, +1.40]). All fold-2 selective contrasts are post hoc.

## S8 Paired bootstrap

Within each dataset×model block, one deterministic schedule of  $B = 2,000$  resamples of image indices is shared across all arms and all conditions, so every contrast is evaluated inside common-image samples. Resamples index evaluated images by position, so a duplicated image contributes its detections and ground truth once per occurrence; this multiplicity is verified by a dedicated unit test (`tests/test_bootstrap_multiplicity.py`), which constructs a stub evaluation where resample  $[0, 0, 1]$  scores 34/101 against the 51/101 that deduplication would give. Reported point estimates are full-sample plug-in values; reported intervals are the 2.5th–97.5th percentiles of the bootstrap distribution of each contrast. Intervals cover image resampling only; they do not cover training-seed, calibration-draw, engine-build, or corruption-realization variance, and image-level resampling treats frames as independent (KITTI frames from shared driving sequences are correlated, so those intervals may be optimistic).

Two-sided bootstrap  $p$ -values use a consistent plus-one convention in each tail,  $\tilde{p} = \min\{1, 2(b_{\min} + 1)/(B + 1)\}$ , so the smallest attainable value at  $B = 2,000$  is  $2/2001 \approx 0.001$  and no  $p$  is ever exactly zero; Holm adjustment covers the nine paired YOLO11 blocks. Heterogeneity of  $\Delta E$  across those blocks uses Cochran’s  $Q = \sum_i w_i (\hat{\Delta}_i - \bar{\Delta}_w)^2$  with  $w_i = 1/\hat{\text{se}}_i^2$  from the bootstrap draws,  $I^2 = \max\{0, (Q - \text{df})/Q\}$ , and the DerSimonian–Laird between-block standard deviation  $\tau$ . Blocks within a dataset share images, so their estimates may be correlated; the cross-block covariance is not identifiable from the phase-B caches because each block’s resample schedule is seeded per dataset×model. To bound the induced dependence we re-ran the nine YOLO11 blocks with  $B = 1,000$  under a schedule seeded per dataset only (`shared.schedule/bootstrap/`): per-draw Pearson correlations of the FP8–INT8  $\Delta E$  between model scales are  $r = \{0.240, 0.213, -0.104\}$  on KITTI ( $n \sim m, n \sim x, m \sim x$ ),  $\{-0.033, -0.066, 0.111\}$  on VOC, and  $\{0.057, 0.013, -0.196\}$  on the COCO subset—small and mixed-sign, so treating the blocks as independent appears only modestly optimistic for the pooled statistics (no generalized  $Q$  with a full covariance matrix was computed) and does not affect the per-block Holm adjustment, which requires valid marginal  $p$ -values but no independence among tests. RetinaNet and intervention intervals are per contrast.

As a cross-machine check, the pretrained-FCOS block was bootstrapped independently on two hosts under the same seed namespace (`coco_fcpos_pretrained`): the 56 cell-level draw arrays agree to  $2.2 \times 10^{-16}$  absolute difference (identical seed,  $B$ , and image set), so resample schedules are portable; container file hashes still differ

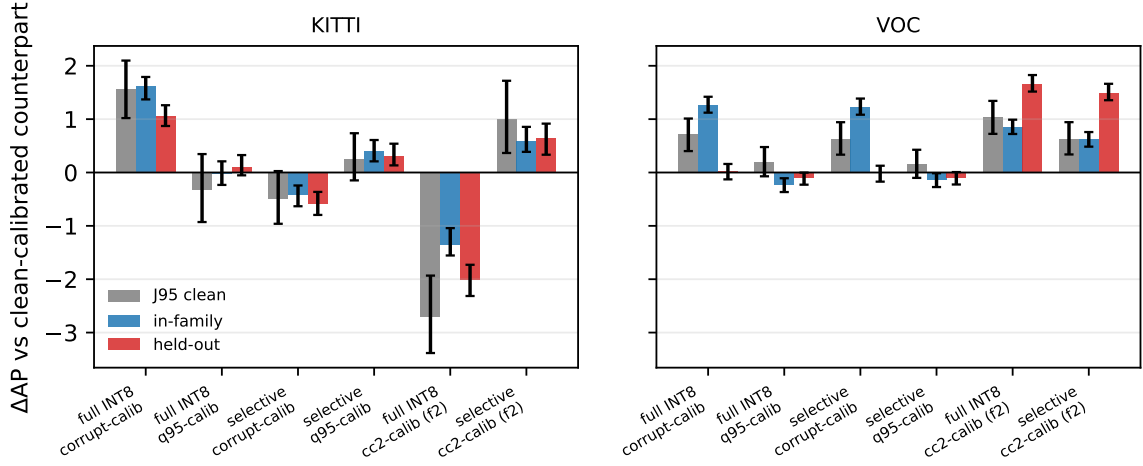


Figure S1: Calibration variants versus their clean-calibrated counterparts (AP points, 95% paired-bootstrap intervals). Grey: J95 clean control; blue: in-family corruptions; red: held-out corruptions; “f2” groups swap the family roles, and for the q95-calibration rows—which see no corrupted images—the in-family/held-out labels mark only the evaluation partition. Held-out bars that merely track the clean bar are not corruption-specific: that is the KITTI fold-1 pattern and, as a harm, the KITTI fold-2 full-INT8 pattern; the VOC fold-2 groups are the exception, where held-out clearly exceeds clean. Per-cell codec effects exist but stay under half an AP point in aggregate—a codec effect on shared failure cells, not evidence of corruption coverage.

because NPZ serialization embeds metadata.

Point estimates are full-sample plug-in values computed from the metric records of each cell (for COCO, from the frozen 2,000-image subset evaluation); bootstrap draws supply intervals, standard errors, and  $p$ -values only. For the intervention, the corruption-specific component on the held-out family is  $\Delta E_{\text{held-out}} = (AP_A - AP_B)_{\text{held-out}} - (AP_A - AP_B)_{J95}$ , the same paired construction as the main estimand. This decomposition is interpretive: it assumes an additive, parallel-shift model in which a calibration-induced level shift on clean inputs transfers comparably to corrupted inputs. Under an interaction model (where the intervention also changes slope across conditions),  $\Delta E_{\text{held-out}}$  mixes that interaction with any clean-level effect; it is not a causal identification theorem. We therefore report the J95-clean and held-out levels alongside it rather than relying on the composite alone.

## S9 Sensitivity analyses

Table S4 resolves the paired  $\Delta E$  of each YOLO11 block per corruption family (each entry is the family-mean FP8–INT8 gap minus the J95 gap) and gives the leave-one-out exclusion of the `motion_blur-s5` severity wall. The sign heterogeneity is also *within* blocks: on VOC-x the JPEG family contributes  $-1.07$  AP while Gaussian noise and motion blur contribute  $+0.69$  and  $+0.67$  AP; on COCO-m JPEG ( $+1.43$ ) and motion blur ( $+0.99$ ) contribute positively while noise contributes a null negative ( $-0.36$ ). Motion blur is the most block-variable family of the four (across-block standard deviation  $\approx 1.1$  AP population,  $\approx 1.2$  sample), with JPEG close behind ( $\approx 1.0$ ); neither is uniformly harmful or helpful. Excluding the severity wall leaves the cross-block heterogeneity essentially unchanged ( $Q \approx 60.2$ ,  $I^2 \approx 87\%$ ) and every Holm-surviving block’s sign intact; the two KITTI blocks that do not survive Holm drift (YOLO11n’s null point estimate flips from  $-0.01$  to  $+0.15$ , and YOLO11m’s unadjusted-significant negative turns null). The interaction pattern is therefore not an artifact of the saturated cell, although the marginal KITTI estimates are sensitive to it.

The same leave-one-out sensitivity applied to the intervention contrasts changes no frozen-criterion verdict; two secondary codec-control contrasts do move across the zero boundary on VOC (both on the held-out basis): INT8-corruptcalib minus INT8-q95calib goes from  $+0.13$   $[-0.02, +0.28]$  to  $+0.21$   $[+0.04, +0.38]$ , and INT8-sel-q95calib minus INT8-selective goes from  $-0.11$   $[-0.22, +0.01]$  to  $-0.15$   $[-0.29, -0.01]$ . Both are read with the main text’s no-multiplicity caveat.

The frozen configuration files are hash-identified: `nn.corruptcalib.v1.20260924.json` SHA-256 `f51011ca5d666b89...` and `nn.corruptcalib.fold2.v1.20260925.json` SHA-256 `5f2d122ae0a52d5f...` (full hashes in the archive manifest).

Freeze timing is established by local-file evidence rather than an external registry: the versioned configuration files’ modification times and the `created_at_utc` fields in the calibration and quantized-ONNX manifests place the configurations before their arms’ first inference (run manifests carry no timestamp field, and the git history records joint archival of configs and outcomes, not pre-inference ordering). The fold-1 file was written 2026-09-25 (its name retains the design date 20260924, and its `title` field says “frozen 2026-09-24”—the title records the design-freeze

day while the serialized file postdates it by one day); the q95 codec-control schedules were added after the fold-1 results were computed—the attempt identifier `nn_q95calib.v1_20260924` keeps the design-date convention and postdates the fold-1 runs in manifest time. The fold-2 primary rule is quoted verbatim: “`int8-cc2calib512 minus int8-matched512 on the fold-2 held-out mean (gaussian_noise+jpeg), paired bootstrap B=2000, same seed namespace as Phase B: CI95 excluding 0 => fold-1 frozen criterion replicated on fold 2`”; we read “replicated” through the directional fold-1 criterion it cites, so a significantly negative change counts as failure.

Table S4: Per-family  $\Delta E$  (AP points; family-mean FP8–INT8 gap minus J95 gap) and the leave-one-out  $\Delta E$  excluding `motion_blur-s5`, per dataset $\times$ model block. \* interval excludes zero.

| Dataset | Model   | Fog    | Gaussian noise | JPEG   | Motion blur | $\Delta E$ excl. mb-s5 |
|---------|---------|--------|----------------|--------|-------------|------------------------|
| KITTI   | YOLO11n | +1.11* | +0.41          | +0.17  | -1.71*      | +0.15 [-0.51, +1.03]   |
| KITTI   | YOLO11m | +0.38  | -0.44          | +0.02  | -2.68*      | -0.35 [-0.98, +0.29]   |
| KITTI   | YOLO11x | -0.40  | -1.98*         | -0.43  | +0.03       | -0.65* [-1.08, -0.29]  |
| VOC     | YOLO11n | +0.35* | +0.28          | +0.27  | +0.34*      | +0.34* [+0.09, +0.62]  |
| VOC     | YOLO11m | -0.03  | -0.48*         | -1.23* | -0.56*      | -0.53* [-0.76, -0.31]  |
| VOC     | YOLO11x | +0.26* | +0.69*         | -1.07* | +0.67*      | +0.13 [-0.06, +0.35]   |
| COCO-2k | YOLO11n | +0.17  | -0.22          | -0.56* | -0.69*      | -0.28 [-0.63, +0.05]   |
| COCO-2k | YOLO11m | +0.36* | -0.36          | +1.43* | +0.99*      | +0.60* [+0.27, +0.97]  |
| COCO-2k | YOLO11x | +0.15  | -0.28          | -2.10* | +0.13       | -0.59* [-1.00, -0.26]  |

## S10 Evidence layers

Diagnostic arms (the RetinaNet decomposition: head-only and all-except-region variants) are single-point recipe sweeps produced during development and carry no intervals. Frozen paired evidence is the 14-condition paired protocol; its freeze covers the estimands, condition schedule, and bootstrap schedule (the nine-block Holm adjustment lives in the analysis code, not the frozen configuration), while the YOLO11 engines and evaluation partitions predate it and are shared with the historical layer. Intervention evidence is the corrupt-calibration experiment together with its codec-matched controls: the two corruption-calibration folds were frozen prospectively before their engines were built, while the fold-1 codec-control arms are post-hoc. Historical exploratory results (a 144-cell grid over four datasets including TT100K, computed on the same engines and estimand under a wider scope and a different analysis plan) are reported only as context and are never pooled with the frozen paired estimates. Post-hoc extensions are the operand-level factorial inside the RetinaNet regression head (W-only and A-only rewires of the frozen matched graph) and the FCOS-R50-FPN replication: both were designed and executed after the frozen paired results were known, are evaluated on the same condition schedule with the same paired bootstrap estimand, and are reported as post-hoc extensions rather than frozen evidence. The FCOS phase package ships its own provenance (per-engine build manifests, quantized-ONNX manifests, per-run training manifests, and the decoder-validation record above) under `phase_h_fcoss_results/`. The pretrained-checkpoint replication (torchvision RetinaNet-R50-FPN-v2 and FCOS-R50-FPN on a paired 2,000-image COCO val2017 subset) is likewise post hoc: its package ships quantized-ONNX manifests, TensorRT build manifests, per-cell input/prediction/metric records, and bootstrap draw caches under `phase_i_coco_pretrained_results/`; calibration used the hash-bound 512-image COCO train2017 manifest `coco_train_clean_512_s20260807_p0_v1_pilotroot` (canonical SHA-256 `bcac5aaa8b29...`). A final layer covers deployment-side audits and post-hoc diagnostics outside the paired estimand: the max-calibration counterfactual pair (`phase_j_maxcalib_results/`), the head-activation captures (`nn_actstats/`), the fivefold engine-rebuild ledger (`nn_rebuild_variance.v1_20260930/`), and the idle-gated latency ledgers (`nn_latency.v1_20260930/`)—single builds and point estimates, no bootstrap intervals.

## S11 Artifact and audit boundary

Manuscript tables and figures are generated deterministically from frozen ledgers by `analysis/build_nn_tables.py`. Engine registries, calibration manifests (including the corrupt-calibration cell-assignment schedule), input manifests, prediction manifests, metric records, and paired-bootstrap draw caches are SHA-256 hash-bound and retained in the accompanying compact-evidence package, deposited in the versioned research archive as release v3.0.3 (version DOI <https://doi.org/10.5281/zenodo.23082850>) under the concept DOI <https://doi.org/10.5281/zenodo.22031663>. Raw benchmark datasets, trained checkpoints, built TensorRT engines, and raw predictions are not redistributed; their provenance and regeneration path is documented in the archive. One binding asymmetry is disclosed for transparency: corrupted, codec-control, and YOLO11 original-clean cells bind each run to a content-addressed input manifest covering the encoded bytes (the YOLO11 clean runs bind the per-image clean

manifests such as `kitti_val_clean_ultralytics.v1`), whereas the RetinaNet original-clean cells bind the frozen dataset split through its annotation SHA-256 (`clean-root:<hash>`). The matching per-image clean manifests exist in the archive and establish the reference bytes of each frozen split, but they are not bound into the RetinaNet clean run records; those runs therefore do not carry an execution-time byte-level binding for the clean inputs, only the annotation-level identifier.

## S12 Deployment-side audits: latency and engine-build determinism

Two post-protocol audits measure deployment-side properties that the frozen accuracy contract does not cover. Both ran on the same RTX 5090 host and TensorRT 11.1.0.106 toolchain as the frozen engines and are retained with hash-bound provenance under `nn_latency.v1.20260930/` and `nn_rebuild_variance.v1.20260930/` in the evidence package. They are descriptive measurements on one host and enter no paired estimand.

**Latency.** Each of the 26 audited engines was timed in-process with CUDA events around `execute_async.v3` at batch 1 (200 warm-up plus 500 timed iterations per repetition, three repetitions; the harness enumerates GPU compute processes before every repetition and refuses to run while a non-allow-listed process holds the device). Table S5 reports per-condition medians and the across-repetition spread; the within-repetition IQR is at most 0.04 ms in every condition. Under this harness the INT8-matched engines run at 1.16–1.31 ms per image, the FP32 references that exist at 4.62–5.09 ms per image (a 3.9–4.0 $\times$  speedup), and the selective-precision repair adds 1.3–6.4% over INT8-matched within the same detector–dataset pair. These are single-host medians: they bound the repair’s deployment cost in the tested setting and do not transfer to other hardware, batch sizes, or power caps.

Table S5: Idle-gated batch-1 latency per engine (median of three repetitions, each the median of 500 timed iterations).

| Detector               | Dataset | Arm               | Median latency (ms) | Rep spread (ms) |
|------------------------|---------|-------------------|---------------------|-----------------|
| FCOS (pretrained)      | COCO    | fp32              | 4.708               | 0.002           |
| FCOS (pretrained)      | COCO    | fp8-matched512    | 1.368               | 0.024           |
| FCOS (pretrained)      | COCO    | int8-matched512   | 1.179               | 0.017           |
| FCOS (pretrained)      | COCO    | int8-selective512 | 1.224               | 0.007           |
| RetinaNet (pretrained) | COCO    | fp32              | 5.087               | 0.055           |
| RetinaNet (pretrained) | COCO    | fp8-matched512    | 1.494               | 0.005           |
| RetinaNet (pretrained) | COCO    | int8-matched512   | 1.307               | 0.012           |
| RetinaNet (pretrained) | COCO    | int8-selective512 | 1.325               | 0.016           |
| FCOS                   | KITTI   | fp32              | 4.615               | 0.054           |
| FCOS                   | KITTI   | fp8-matched512    | 1.356               | 0.033           |
| FCOS                   | KITTI   | int8-matched512   | 1.197               | 0.047           |
| FCOS                   | KITTI   | int8-selective512 | 1.219               | 0.017           |
| RetinaNet              | KITTI   | fp8-matched512    | 1.359               | 0.056           |
| RetinaNet              | KITTI   | int8-aonly512     | 1.188               | 0.012           |
| RetinaNet              | KITTI   | int8-matched512   | 1.171               | 0.025           |
| RetinaNet              | KITTI   | int8-selective512 | 1.225               | 0.033           |
| RetinaNet              | KITTI   | int8-wonly512     | 1.188               | 0.017           |
| FCOS                   | VOC     | fp32              | 4.618               | 0.002           |
| FCOS                   | VOC     | fp8-matched512    | 1.365               | 0.035           |
| FCOS                   | VOC     | int8-matched512   | 1.162               | 0.011           |
| FCOS                   | VOC     | int8-selective512 | 1.236               | 0.036           |
| RetinaNet              | VOC     | fp8-matched512    | 1.350               | 0.022           |
| RetinaNet              | VOC     | int8-aonly512     | 1.215               | 0.003           |
| RetinaNet              | VOC     | int8-matched512   | 1.192               | 0.040           |
| RetinaNet              | VOC     | int8-selective512 | 1.247               | 0.037           |
| RetinaNet              | VOC     | int8-wonly512     | 1.221               | 0.004           |

**Engine-build determinism.** Rebuilding each of seven engines five times from the same quantized ONNX under the frozen build policy (TF32 disabled, optimization level 3, 4096 MiB workspace) produced a distinct engine SHA-256 and a distinct inspector SHA-256 in all 35 builds, and the fused-layer count itself varies across rebuilds (270–309 layers for the same graph; Table S6): under this builder the tactic autotuner does not settle on a bit-identical plan, or even the same fusion boundaries. The nondeterminism does not reach the metric in the evaluated slice—across five rebuilds of the KITTI INT8-matched engine the measured AP is identical on both the clean and codec-control bases (spread 0.0000 AP)—but the evaluation coverage is one arm on one dataset, so the audit bounds rather than eliminates build-to-build metric variance.

## S13 Head-region activation statistics

To probe the operand-factorial attribution at the distribution level we captured the activation tensors consumed by each RetinaNet and FCOS head on frozen 512-image clean-calibration manifests, running the unquantized ONNX graph in ONNX-Runtime (CUDA provider) with every head-consumed tensor exposed as a graph output. Per-channel  $|x|_{\max}$  and excess kurtosis are computed over all spatial positions in the 512 captured images, with

Table S6: Fivefold rebuild audit per arm: every rebuild is bit-distinct at both engine and inspector level; AP spread is reported where evaluated.

| Arm                               | $n$ | distinct eng. SHA | distinct insp. SHA | layers  | AP spread      |
|-----------------------------------|-----|-------------------|--------------------|---------|----------------|
| coco_retinanet_int8-matched512    | 5   | 5                 | 5                  | 285–309 | –              |
| coco_retinanet_int8-selective512  | 5   | 5                 | 5                  | 279–294 | –              |
| kitti_retinanet_int8-aonly512     | 5   | 5                 | 5                  | 270–274 | –              |
| kitti_retinanet_int8-matched512   | 5   | 5                 | 5                  | 279–298 | 0.0000; 0.0000 |
| kitti_retinanet_int8-selective512 | 5   | 5                 | 5                  | 270–287 | 0.0000; 0.0000 |
| voc_retinanet_int8-matched512     | 5   | 5                 | 5                  | 277–282 | –              |
| voc_retinanet_int8-selective512   | 5   | 5                 | 5                  | 277–286 | –              |

moments accumulated by streaming over every pixel. Each tensor’s per-channel statistics are first summarized by their median, then the regional and per-level entries report medians across tensors. Each tensor is assigned to a region by graph consumption (tensors feeding both heads are “shared head inputs”). Captures cover the KITTI and VOC trained RetinaNet and FCOS graphs and the COCO pretrained RetinaNet graph (`nn_actstats/` in the evidence package; input bytes hash-verified against the calibration manifests).

Table S7 reports the summaries. On the three RetinaNet graphs the regression head is *not* uniformly larger in typical magnitude—its median channel amplitude is comparable to the classification head’s—but it is markedly heavier-tailed (on the KITTI graph, median excess kurtosis 10.6 versus 3.1 for the classification head, with per-channel maxima reaching 204795 versus 21826; on VOC, 8.0 versus 2.5; on the COCO pretrained graph, 6.0 versus 2.8). The tail is broad rather than a single stray channel: the median of the per-tensor channel-p95 excess kurtosis is 43.6 versus 9.1 on KITTI and 52.9 versus 6.7 on VOC. Rare large activations inflate a single per-tensor scale and spend the INT8 code budget on outlier coverage, which is exactly the signature under which activation—not weight—quantization dominates the deficit (the operand factorial of Section 3.7 of the main text). Two cautions bound the reading: these are activations of the fp32 ONNX graph on calibration inputs, not the deployed quantizer’s scale values (the calibrator’s chosen activation amaxes on the matched INT8 graphs are recorded alongside, e.g. a median of 25.3 versus 12.5 on KITTI); and the captures are a measurement of the calibration distribution, which is itself what the quantizer saw—the tails under corrupted inputs are not measured here.

Table S8 resolves the same captures per FPN level, and a structural audit of the matched QDQ graphs (`nn_actstats/qdq_structure*.json`) records how the head’s activation ranges are parametrized. Two structure-level facts matter. First, the regression head does not share one activation scale across pyramid levels: the matched graph carries 25 regression-head-consumed activation quantizers—the five FPN feature tensors at the head input boundary (shared with the classification head) plus four per-level tensors in each of the five level-unrolled tower stages—and every one is per-tensor with a level-specific scale, so a shared head-wide activation range is not the mechanism here. Second, the tail burden is level-ordered: regression-head median excess kurtosis is highest at the fine levels that carry small-object predictions (427.5 at P3 on KITTI, falling to 2.8 at P7; the classification head shows a much flatter profile, 20.1 to 0.1), and the same ordering holds on VOC and on the pretrained COCO graph. Per-tensor grids at P3/P4 therefore spend the most code budget covering outliers—consistent with the matched arm’s  $AP_S/AP_{75}$ -concentrated deficit in Table 8 of the main text—though this is a consistency argument, not a causal isolation.

The FCOS captures provide a cross-architecture falsification check for the extreme-tail profile, not evidence that FCOS activations are tail-free. Regression-head median excess kurtosis at P3/P4 is 14.4/11.3 on KITTI and 3.8/4.7 on VOC, compared with 427.5/208.1 and 179.1/143.0 for RetinaNet. In particular, FCOS still has positive excess kurtosis at KITTI P3; its fine-level tails are substantially lighter, not absent. This co-occurrence—lighter fine-level tails and no comparable head-specific deficit—is consistent with the proposed distribution-level account, but FCOS also differs in architecture, anchor density, centerness scoring, and selective-output coverage, so the comparison does not isolate tail shape causally.

## S14 Max-calibration counterfactual

Section S13 shows the regression-head activations are heavy-tailed under the entropy calibrator’s inputs, which admits two distinct cost accounts: tail clipping (the chosen range excludes outliers) or bin resolution (the range is fine but 127 levels spread too coarsely across it). A max-calibration arm separates the two: it sets each activation scale to the absolute maximum observed on the calibration set, removing calibration-set clipping at the price of the coarsest bin resolution (evaluation-time activations beyond that maximum can still clip). We therefore ran a post-hoc counterfactual pair—INT8-matched and INT8-selective under the max estimator, same graphs, same frozen 512-image clean-calibration manifests, same 14-condition evaluation schedule—on the KITTI and VOC RetinaNet detectors (`nn_maxcalib_v1_20260930`; single engine builds, point estimates only, no bootstrap draws).

Table S9 reports the levels. The counterfactual is sharp and negative in both directions. On KITTI, max calibration does not repair the matched deficit—it deepens it, from 28.29 to 14.02 AP on the clean basis—while the head-exclusion benefit persists under the new estimator (selective minus matched 9.47 AP clean, 5.51 AP corrupt-

Table S7: Per-region activation statistics over the 512-image clean calibration sets for trained RetinaNet and FCOS graphs and the pretrained COCO RetinaNet graph.  $|x|_{\max}$  is the per-channel absolute maximum; excess kurtosis is Fisher. For each tensor, channel-level statistics are summarized by a median; table medians are then taken across tensors. “Shared head inputs” are tensors consumed by both heads (FPN boundary features).

| Graph                       | Region              | $n$ tensors | med. $ x _{\max}$ | max $ x $ | med. excess kurt. | max ex. kurt. |
|-----------------------------|---------------------|-------------|-------------------|-----------|-------------------|---------------|
| KITTI RetinaNet (trained)   | regression head     | 192         | 8.60              | 7491      | 10.63             | 204795        |
|                             | classification head | 192         | 10.32             | 6156      | 3.14              | 21826         |
|                             | shared head inputs  | 5           | 48.26             | 524       | 16.54             | 354           |
| VOC RetinaNet (trained)     | regression head     | 192         | 9.34              | 2290      | 7.99              | 261078        |
|                             | classification head | 192         | 8.34              | 1116      | 2.52              | 62412         |
|                             | shared head inputs  | 5           | 74.33             | 576       | 5.87              | 96            |
| COCO RetinaNet (pretrained) | regression head     | 192         | 7.57              | 1282      | 6.04              | 313956        |
|                             | classification head | 192         | 7.52              | 815       | 2.79              | 21296         |
|                             | shared head inputs  | 5           | 50.92             | 270       | 3.06              | 43            |
| KITTI FCOS (trained)        | regression head     | 217         | 2.22              | 49        | 1.17              | 59256         |
|                             | classification head | 192         | 5.29              | 89        | 1.39              | 84352         |
|                             | shared head inputs  | 5           | 1.18              | 5         | 10.29             | 61            |
| VOC FCOS (trained)          | regression head     | 217         | 2.45              | 183       | 1.44              | 373590        |
|                             | classification head | 192         | 7.61              | 200       | 1.81              | 3276795       |
|                             | shared head inputs  | 5           | 9.37              | 54        | 5.82              | 26            |

Table S8: Per-FPN-level activation statistics, reported as the median of per-tensor median-channel  $|x|_{\max}$  / excess kurtosis at each level (P3 finest, P7 coarsest; shared FPN boundary features are included where applicable). The table includes three RetinaNet and two FCOS graphs; fine-level tails are substantially heavier in RetinaNet.

| Graph                       | Head | P3       | P4       | P5      | P6     | P7     |
|-----------------------------|------|----------|----------|---------|--------|--------|
| KITTI RetinaNet             | reg. | 22.6/427 | 14.6/208 | 10.5/81 | 6.7/8  | 3.6/3  |
|                             | cls. | 19.6/20  | 15.8/19  | 10.2/9  | 6.3/2  | 3.5/0  |
| VOC RetinaNet               | reg. | 29.9/179 | 18.5/143 | 10.5/76 | 7.1/28 | 4.6/8  |
|                             | cls. | 11.6/4   | 9.9/5    | 10.2/6  | 7.3/3  | 4.7/1  |
| COCO RetinaNet (pretrained) | reg. | 21.3/61  | 14.1/77  | 9.2/41  | 6.1/20 | 4.1/8  |
|                             | cls. | 10.6/3   | 9.2/4    | 7.7/4   | 6.0/2  | 4.2/1  |
| KITTI FCOS                  | reg. | 5.2/14   | 3.4/11   | 2.3/6   | 1.0/0  | 0.6/-0 |
|                             | cls. | 13.3/11  | 10.9/9   | 7.5/3   | 3.2/0  | 2.2/-1 |
| VOC FCOS                    | reg. | 3.3/4    | 3.3/5    | 2.7/5   | 1.8/2  | 0.8/-0 |
|                             | cls. | 10.9/4   | 11.5/6   | 9.3/6   | 6.3/3  | 2.4/-1 |

mean), so the regression-head sensitivity is not an artifact of the entropy estimator. On VOC the max estimator is catastrophic on *both* arms (J95 clean  $\approx 0.001$  AP): the detection-score distribution collapses (the 99th-percentile detection score on the clean run drops from  $\approx 0.91$  under entropy calibration to  $\approx 0.15$ ; `voc_score_audit.json` in the evidence package), consistent with a failure in the score (classification) pathway that regression-head exclusion cannot rescue. A 128-image draw of the same estimator in the earlier recipe sweep scored 23.94 AP on the KITTI clean basis—the max estimator’s outcome is itself draw-sensitive, since a single unlucky outlier image inflates every range.

Because the graph-wide swap perturbs every quantizer, it cannot separate a head-local estimator effect from collateral damage elsewhere (the VOC collapse is the extreme case). We therefore ran a second, head-local counterfactual (`nn_maxreg.v1.20261001`; same single-build, point-estimate-only protocol): starting from the frozen INT8-matched graphs, the entropy-calibrated scales of the regression-head activation quantizers (25 per graph, per-tensor, one per conv per FPN level; Section S13) were replaced by the calibration-set maxima measured on the same captured activations, while every other quantizer—backbone, classification head, head-input boundary—kept its entropy-calibrated scale byte-identical. The head-local arm leaves VOC functional and still does not repair the deficit (Table S9): clean J95 sits at 27.15 AP on KITTI and 52.32 AP on VOC, against 28.29/53.30 for the entropy-matched arms, and the corrupt-mean levels (19.58/37.93 AP) track the entropy-matched arms within about one AP point. Widening only the regression-head ranges to the observed maxima recovers nothing.

The reading we take is bounded: clipping of the observed calibration range is not the operative cost, because widening the range to the calibration maximum—graph-wide or restricted to the regression head—does not recover accuracy and can destroy the classifier outright. That leaves the resolution side of the account—a fixed 127-level grid over a wide range—as the better candidate, though we did not separate resolution from the other numerical changes a coarser scale induces (larger rounding bias on in-range values, dequantization shift), and max calibration does not preclude evaluation-time clipping of activations beyond the calibration-set maximum. Corruption-aware calibration and max calibration together bracket the range-choice dimension: neither repairs the deficit, and the regression-head benefit survives both.

Table S9: Max-calibration counterfactual on RetinaNet (post-hoc diagnostic, single builds, plug-in AP points without intervals). Reference rows repeat the frozen entropy-calibrated paired-protocol arms for the same detector–dataset pairs; bold rows are the max-estimator arms on the same 14-condition schedule: graph-wide matched/selective, then the head-local variant with the max estimator applied to regression-head activation quantizers only.

| Dataset | Arm                                      | J95 clean | codec ctrl. | corrupt mean |
|---------|------------------------------------------|-----------|-------------|--------------|
| KITTI   | FP32 reference                           | 50.15     | –           | 37.80        |
|         | FP8-matched (entropy)                    | 49.48     | 49.25       | 36.75        |
|         | INT8-matched (entropy)                   | 28.29     | 28.18       | 20.37        |
|         | INT8-selective (entropy)                 | 49.60     | 49.53       | 36.58        |
|         | <b>INT8-matched (max)</b>                | 14.017    | 14.176      | 7.817        |
|         | <b>INT8-selective (max)</b>              | 23.484    | 24.072      | 13.326       |
|         | <b>INT8-matched (max, reg head only)</b> | 27.148    | 27.429      | 19.584       |
| VOC     | FP32 reference                           | 57.23     | –           | 42.32        |
|         | FP8-matched (entropy)                    | 56.17     | 56.21       | 41.48        |
|         | INT8-matched (entropy)                   | 53.30     | 53.18       | 38.71        |
|         | INT8-selective (entropy)                 | 55.87     | 55.79       | 40.52        |
|         | <b>INT8-matched (max)</b>                | 0.001     | 0.002       | 0.005        |
|         | <b>INT8-selective (max)</b>              | 0.002     | 0.002       | 0.005        |
|         | <b>INT8-matched (max, reg head only)</b> | 52.316    | 52.162      | 37.929       |

## S15 KITTI final-holdout check

As a post-hoc check of the RetinaNet head-localization result, we evaluated the four factorial arms on the 1,197-image KITTI final-holdout partition reserved at split time. The partition was excluded from model and recipe development, but this re-evaluation was not a preregistered test; no retraining or calibration changes were made. The same engines, decoder, and 14-condition evaluation schedule were used (`nn_holdout_v1_20261001`); the retained hash-bound metric and input manifests identify the evaluated artifacts. Values are full-sample point estimates only, with no bootstrap intervals. The selection-partition rows are included as a descriptive reference, not as a paired comparison to the holdout rows. Since the split is frame-level, adjacent frames from the same driving sequence can cross partitions; this is not a sequence-independent generalization test.

On the holdout, selective and W-only recover the clean-level deficit, while A-only remains near the matched arm. The pattern therefore reproduces the activation-dominant operand attribution observed on the selection partition, although the absolute AP levels differ between image samples.

Table S10: KITTI RetinaNet factorial arms on the final holdout and, for context, the selection partition. The holdout contains 1,197 images reserved at split time; values are plug-in AP points without intervals.

| Split     | Arm            | J95 clean | codec ctrl. | corrupt mean |
|-----------|----------------|-----------|-------------|--------------|
| Holdout   | INT8-matched   | 34.70     | 34.42       | 24.83        |
| Holdout   | INT8-selective | 65.25     | 65.17       | 46.01        |
| Holdout   | INT8 W-only    | 65.30     | 64.90       | 45.99        |
| Holdout   | INT8 A-only    | 34.64     | 34.71       | 24.86        |
| Selection | INT8-matched   | 28.29     | 28.18       | 20.37        |
| Selection | INT8-selective | 49.60     | 49.53       | 36.58        |
| Selection | INT8 W-only    | 49.50     | 49.44       | 36.52        |
| Selection | INT8 A-only    | 28.31     | 28.14       | 20.36        |

## S16 Minimum reporting checklist for paired corruption studies

- record the complete treatment recipe (checkpoint, quantizer coverage, calibration list and preprocessing, builder settings, decoder), not just the datatype label;
- pair treatments on identical image bytes, including terminal encoding;
- subtract the matched clean gap before reading a corruption interaction;
- report absolute corrupted AP and the clean gap alongside every interaction, so floor effects cannot masquerade as robustness;
- keep calibration corruption families disjoint from held-out evaluation families when claiming calibration-based robustness;
- carry codec-matched calibration controls before attributing calibration-intervention gains to corruption coverage;
- flag severity-wall cells where every treatment including FP32 is at the floor, and report a leave-one-out sensitivity;
- audit engine-build determinism before treating a serialized engine as a fixed artifact—tactic autotuning makes rebuilds bit-distinct, so hash the engine and inspector and spot-check metric spread;
- gate latency claims on an idle-device harness with repetitions and dispersion, and scope them to the measured host.